

Oct. 3rd.

Date.

The inferior limits of a bounded sequence are the smallest partial limits of the sequence.

Proof: let $i = \liminf_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \inf_{k \geq n} x_k$

$\forall \varepsilon > 0, \exists N \in \mathbb{N}$, s.t. $i_N > i - \varepsilon$

$\exists k$, when $k > N$, $x_k > i_N$, then $x_k > i - \varepsilon$.

then $x_k > i - \varepsilon$, i.e. $\lim_{k \rightarrow \infty} x_k > i$

By the definition of infimum

$i_1 \leq x_{n_1} < i_1 + 1$, $i_2 \leq x_{n_2} < i_2 + 1$

then $i_{k+1} \leq x_{n_{k+1}} < i_{k+1} + \frac{1}{k}$, thus $\lim_{k \rightarrow \infty} x_{n_k} = i$. (By the squeeze theorem)

Oct. 5th.

82. $x_n = a_0 + a_1 q + \dots + a_n q^n$. $\# \mid a_k \mid < M (k=0, 1, 2, \dots)$ & $\mid q \mid < 1$

$\forall m, n \in \mathbb{N}$, w.l.g. $m > n$.

$$\mid x_m - x_n \mid = \mid a_{n+1} q^{n+1} + a_{n+2} q^{n+2} + \dots + a_m q^m \mid$$

Since $\mid q \mid < 1$.

$$\leq \mid (m-n) M (q^{n+1} + q^{n+2} + \dots + q^m) \mid$$

$$\lim_{n \rightarrow \infty} \mid q \mid^{n+1} = 0$$

$$= \mid (m-n) M \cdot \frac{q^{n+1} (1 - q^{m-n})}{1 - q} \mid$$

$\forall \varepsilon > 0, \exists N \in \mathbb{N}$, for any $m, n > N$ we have $\mid x_m - x_n \mid < \varepsilon$.

$$< (m-n) M \cdot \frac{\mid q \mid^{n+1}}{1 - \mid q \mid}$$

Hence, the sequence is Cauchy's.

$$83. x_n = \frac{\sin 1}{2} + \frac{\sin 2}{2^2} + \dots + \frac{\sin n}{2^n}$$

$$\text{since } \lim_{n \rightarrow \infty} \frac{1}{2^n} = 0$$

$\forall m, n \in \mathbb{N}$, w.l.g. $m > n$.

$\forall \varepsilon > 0$, let $N = \lceil \frac{\ln \varepsilon}{\ln \frac{1}{2}} \rceil$, for any $m, n > N$ we have $\mid x_m - x_n \mid < \varepsilon$

$$\mid x_m - x_n \mid = \left| \frac{\sin(n+1)}{2^{n+1}} + \frac{\sin(n+2)}{2^{n+2}} + \dots + \frac{\sin m}{2^m} \right|$$

Hence, the sequence is Cauchy's

$$\leq \left| \frac{1}{2^{n+1}} + \dots + \frac{1}{2^m} \right|$$

$$< \frac{1}{2^n}$$

$$84. x_n = \frac{\cos 1!}{1 \cdot 2} + \frac{\cos 2!}{2 \cdot 3} + \dots + \frac{\cos n!}{n(n+1)}.$$

$\forall m, n \in \mathbb{N}$, w.l.g. $m > n$.

$$|x_m - x_n| = \left| \frac{\cos(n+1)!}{(n+1)(n+2)} + \frac{\cos(n+2)!}{(n+2)(n+3)} + \dots + \frac{\cos m!}{m(m+1)} \right|$$

$$< \left| \frac{1}{(n+1)(n+2)} + \frac{1}{(n+2)(n+3)} + \dots + \frac{1}{m(m+1)} \right| < \left| \frac{1}{n+1} - \frac{1}{m+1} \right| < \frac{1}{n+1}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n+1} = 0.$$

$\forall \varepsilon > 0$, let $N = \lceil \frac{1}{\varepsilon} - 1 \rceil$, For any $m, n > N$, we have $|x_m - x_n| < \varepsilon$.

Hence, the sequence is convergent.

$$85. x_n = 1 + \frac{1}{2^2} + \dots + \frac{1}{n^2}$$

~~For~~ $\forall m, n \in \mathbb{N}$, w.l.g. $m > n$

$$|x_m - x_n| = \left| \frac{1}{(n+1)^2} + \frac{1}{(n+2)^2} + \dots + \frac{1}{m^2} \right|$$

$$< \left| \frac{1}{n(n+1)} + \dots + \frac{1}{m(m-1)} \right|$$

$$< \frac{1}{n}.$$

similar as (84), the sequence is convergent.

87. $\exists \varepsilon_0 > 0$, For any $N \in \mathbb{N}$, $\exists (m > N) \wedge (n > N)$,

$$\text{s.t. } |x_m - x_n| \geq \varepsilon_0.$$

88. Proof: Let $m=2n$ ($n \in \mathbb{N}$)
 $|x_m - x_n| = \left| \frac{1}{n^2} + \dots + \frac{1}{n^2} \right| > n \cdot \frac{1}{n^2} = \frac{1}{n}$

Let $\varepsilon_0 = \frac{1}{2}$. For any $N \in \mathbb{N}$, we can find $n > N$,
 such that $|x_m - x_n| > \frac{1}{n} > \varepsilon_0 = \frac{1}{2}$. Hence the sequence is divergent.

89. Verify: If the sequence $\{x_n\}$ is convergent, then any subsequence
 of $\{x_n\}$, we denote by $\{x_{p_n}\}$ is convergent, and $\lim_{n \rightarrow \infty} x_{p_n} = \lim_{n \rightarrow \infty} x_n$

Proof: w.l.g. $\lim_{n \rightarrow \infty} x_n = A$.

$\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$, for any $n > N$, $|x_n - A| < \varepsilon$

Since $\lim_{n \rightarrow \infty} p_n = \infty$, we can find $p_n > N$, s.t. $|x_{p_n} - A| < \varepsilon$

90. Verify: If a monotonous sequence has a monotonous
 subsequence, then the sequence is convergent

Proof: w.l.g. $\lim_{n \rightarrow \infty} x_{p_n} = A$. Suppose $\{x_n\}$ is increasing

$\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$, $K > N$ $|x_{p_K} - A| < \varepsilon$

Since $\lim_{n \rightarrow \infty} p_n = \infty$, we can find $p_K \leq n < p_{K+1}$

$|x_{p_K} - A| < \varepsilon$ and $|x_{p_{K+1}} - A| < \varepsilon$ then $|x_n - A| < \varepsilon$,

Hence, $\lim_{n \rightarrow \infty} x_n = A$.

91. Verify: If $\lim_{n \rightarrow \infty} x_n = a$, then $\lim_{n \rightarrow \infty} |x_n| = |a|$

$\forall \varepsilon > 0, \exists N \in \mathbb{N}$, for any $n > N$

$$|x_n - a| < \varepsilon.$$

$$||x_n| - |a|| \leq |x_n - a| < \varepsilon. \text{ Hence } \lim_{n \rightarrow \infty} |x_n| = |a|$$

92. Let $\lim_{n \rightarrow \infty} x_n = a$, $\lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = \underline{\hspace{2cm}}$

1° $a \neq 0$, $\lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = \frac{\lim_{n \rightarrow \infty} x_{n+1}}{\lim_{n \rightarrow \infty} x_n} = 1.$

2° $a = 0$ For the sequence $1, 1, \frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{1}{3}, \dots, \frac{1}{n}, \frac{1}{n}, \dots$
 $\frac{x_{n+1}}{x_n} = 1 \text{ or } \frac{1}{2}$, does not have limit.

If the limit exists, $\lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = b$ s.t. $|b| \leq 1$.

Proof: Assume $|b| > 1$.

Let $\varepsilon = \frac{|b|-1}{2} > 0$.

$\exists N \in \mathbb{N}$, for any $n > N$, $|\frac{x_{n+1}}{x_n}| > \varepsilon$.

$$|x_n| = |x_N| \cdot |\frac{x_{N+1}}{x_N}| \cdots |\frac{x_n}{x_{n-1}}| > |x_N| \cdot \varepsilon^{n-N}.$$

$\varepsilon > 1$, $\lim_{n \rightarrow \infty} \varepsilon^{n-N} = +\infty$, which contradicts with $\lim_{n \rightarrow \infty} x_n = a$

93. Verify: convergent sequence is bounded.

Proof: Let $\lim_{n \rightarrow \infty} x_n = a$.

let $\varepsilon_0 = 1$. $\exists N \in \mathbb{N}$, $n > N$, $|x_n - a| < 1$, $\Rightarrow |x_n| < |a| + 1$

Let $M = \max\{|x_1|, |x_2|, \dots, |x_N|, |a| + 1\}$, then $|x_n| \leq M$

Hence, $\{x_n\}$ is bounded.

95. Verify: $x_n \rightarrow +\infty$ ($n=1, 2, \dots$) is bound to reach its infimum.

~~Let $M > 0$, $\exists N \in \mathbb{N}$, $n > N$, s.t. $|x_n| \geq M$~~

Let $\forall M > 0$, $\exists N \in \mathbb{N}$, $n > N$, s.t. $|x_n| \geq M$.

Then the $\inf x_n = \min \{x_1, x_2, \dots, x_N\}$, thus, $\inf x_n \in \{x_n\}$.

96. x_n ($n=1, 2, \dots$). $\inf \{x_n\}$, $\sup \{x_n\}$, $\lim_{n \rightarrow \infty} x_n$ and $\overline{\lim}_{n \rightarrow \infty} x_n$.

[101]. $x_n = 1 - \frac{1}{n}$.

$$x_{n+1} - x_n = \frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)} > 0$$

$\{x_n\}$ is increasing.

$$\inf \{x_n\} = 0 \quad \sup \{x_n\} = 1. \quad \lim_{n \rightarrow \infty} x_n = 1 \quad \overline{\lim}_{n \rightarrow \infty} x_n = 1$$

$$[102] \quad x_n = \frac{(-1)^n}{n} + \frac{1+(-1)^n}{2}$$

$$\inf \{x_n\} = -1. \quad \sup \{x_n\} = \frac{3}{2}$$

$$\lim_{n \rightarrow \infty} x_n = 0 \quad \overline{\lim}_{n \rightarrow \infty} x_n = 1$$

$$[103] \quad x_n = 1 + \frac{n}{n+1} \cos \frac{n\pi}{2}$$

$$\inf \{x_n\} = 0 \quad \sup \{x_n\} = 2 \quad \lim_{n \rightarrow \infty} x_n = 0 \quad \overline{\lim}_{n \rightarrow \infty} x_n = 2$$

$$[104] \quad x_n = 1 + 2(-1)^{n+1} + 3(-1)^{\frac{n(n-1)}{2}}$$

$$\inf \{x_n\} = -4 \quad \sup \{x_n\} = 6 \quad \lim_{n \rightarrow \infty} x_n = -4 \quad \overline{\lim}_{n \rightarrow \infty} x_n = 6$$

$$[105] \quad x_n = \frac{n-1}{n+1} \cos \frac{2n\pi}{3}$$

$$\inf \{x_n\} = -\frac{1}{2} \quad \sup \{x_n\} = 1 \quad \lim_{n \rightarrow \infty} x_n = -\frac{1}{2} \quad \overline{\lim}_{n \rightarrow \infty} = 1$$

$$[106] \quad x_n = (-1)^n (n)$$

$$\inf \{x_n\} = \lim_{n \rightarrow \infty} x_n = -\infty \quad \sup \{x_n\} = \overline{\lim}_{n \rightarrow \infty} x_n = 1$$

$$[107] \quad x_n = -n[2 + (-1)^n]$$

$$\inf \{x_n\} = \lim_{n \rightarrow \infty} x_n = -\infty$$

$$\sup \{x_n\} = -1$$

$$\overline{\lim}_{n \rightarrow \infty} x_n = -\infty$$

$$[108] \quad x_n = n^{(-1)^n}$$

$$\inf \{x_n\} = \lim_{n \rightarrow \infty} x_n = 0 \quad \sup \{x_n\} = +\infty \quad \overline{\lim}_{n \rightarrow \infty} x_n = +\infty$$

$$[109] \quad x_n = 1 + n \sin \frac{n\pi}{2}$$

$$\inf \{x_n\} = -\infty \quad \lim_{n \rightarrow \infty} x_n = -\infty$$

$$\sup \{x_n\} = +\infty \quad \overline{\lim}_{n \rightarrow \infty} x_n = +\infty$$

$$[110] \quad x = \frac{1}{n-10.2}$$

$$\inf \{x_n\} = -5 \quad \sup \{x_n\} = \frac{5}{4}$$

$$\lim_{n \rightarrow \infty} x_n = 0 \quad \overline{\lim}_{n \rightarrow \infty} x_n = 0$$

Date.

Find $\overline{\lim}_{n \rightarrow \infty} x_n$ and $\lim_{n \rightarrow \infty} x_n$

$$[111] x_n = \frac{n^2}{1+n^2} \cos \frac{2n\pi}{3}$$

$$\lim_{n \rightarrow \infty} x_n = -\frac{1}{2} \quad \overline{\lim}_{n \rightarrow \infty} = 1$$

$$[112] x_n = \left(1 + \frac{1}{n}\right)^n \cdot (-1)^n + \sin \frac{n\pi}{4}$$

$$\lim_{n \rightarrow \infty} x_n = -e - \frac{\sqrt{2}}{2} \quad \overline{\lim}_{n \rightarrow \infty} x_n = e + 1$$

$$[113] x_n = \frac{n}{n+1} \sin^2 \frac{n\pi}{4}$$

$$x_n = \frac{n}{n+1} \left(\frac{1 - \cos \frac{n\pi}{2}}{2} \right)$$

$$\lim_{n \rightarrow \infty} x_n = 0$$

$$\overline{\lim}_{n \rightarrow \infty} x_n = 1$$

$$[114] x_n = \sqrt[n]{1 + 2^{n \cdot (-1)^n}}$$

$$1^\circ n = 2k (k \in \mathbb{N}), \quad x_n = \sqrt[n]{1 + 2^n} \quad \lim_{n \rightarrow \infty} x_n = 1$$

$$2^\circ n = 2k+1 (k \in \mathbb{N}), \quad x_n = \sqrt[n]{1 + 2^n}$$

$$(2^n)^{\frac{1}{n}} < (1 + 2^n)^{\frac{1}{n}} < (2^{n+1})^{\frac{1}{n}}$$

$$\lim_{n \rightarrow \infty} (2^n)^{\frac{1}{n}} = 2$$

$$\lim_{n \rightarrow \infty} 2^{\frac{n+1}{n}} = e^{\ln 2 \cdot \lim_{n \rightarrow \infty} \frac{n+1}{n}} = e^{\ln 2} = 2$$

By the squeeze theorem, $\lim_{n \rightarrow \infty} x_n = 2$. i.e. $\overline{\lim}_{n \rightarrow \infty} x_n = 2$.

$$[115] \quad x_n = \cos^n \frac{2n\pi}{3}$$

$$\lim_{n \rightarrow \infty} x_n = 0$$

$$\lim_{n \rightarrow \infty} x_n = 1$$

Find the limit point

$$[116] \quad \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{1}{8}, \frac{7}{8}, \dots, \frac{1}{2^n}, \frac{2^n-1}{2^n}, \dots$$

The sequence has limit points 0 and 1

$$[117] \quad 1, \frac{1}{2}, 1+\frac{1}{2}, \frac{1}{3}, 1+\frac{1}{3}, \frac{1}{2}+\frac{1}{3}, \frac{1}{4}, 1+\frac{1}{4}, \dots, \frac{1}{n}, 1+\frac{1}{n}, \dots$$

We have subsequences $\{1+\frac{1}{n}\}$, $\{\frac{1}{n}\}$, $\{\frac{1}{2}+\frac{1}{n}\}$, $\{\frac{1}{3}+\frac{1}{n}\}$, $\{\frac{1}{n-1}+\frac{1}{n}\}$, $\{\frac{1}{n}\}$, $\dots$

So it has limit points 0, 1, $\frac{1}{2}$, $\frac{1}{3}$, $\dots$, $\frac{1}{n}$, $\dots$

$$[118] \quad \frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}, \dots$$

Every point of the close interval $[0, 1]$ is the limit point of it.

$$[119] \quad x_n = 3(1 - \frac{1}{n}) + 2(-1)^n$$

$\lim_{n \rightarrow \infty} 3(1 - \frac{1}{n}) = 3$, the limit points are 5 and 1

$$[120] \quad x_n = \frac{1}{2} [(a+b) + (-1)^n (a-b)]$$

$$x_{2k} = a \quad x_{2k-1} = b$$

the limit points, a, b.

$$[2552] \sum_{n=1}^{\infty} (\sqrt{n+2} - 2\sqrt{n+1} + \sqrt{n})$$

$$a_n = (\sqrt{n+2} - \sqrt{n+1}) - (\sqrt{n+1} - \sqrt{n})$$

$$S_n = \sqrt{n+2} - \sqrt{n+1} - (\sqrt{2} - 1) = 1 - \sqrt{2} + \frac{1}{\sqrt{n+2} + \sqrt{n+1}}$$

since $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+2} + \sqrt{n+1}} = 0$. then $\lim_{n \rightarrow \infty} S_n = 1 - \sqrt{2}$.

$$[2553] \sum_{n=1}^{\infty} \sin nx.$$

1° $x = k\pi, k \in \mathbb{Z}, \sin nx \equiv 0$. $\therefore \sum_{n=1}^{\infty} \sin nx = 0$.

2° $x \neq k\pi, k \in \mathbb{Z}$, we now ~~sh~~ should verify $\lim_{n \rightarrow \infty} \sin nx \neq 0$.

Assume $\lim_{n \rightarrow \infty} \sin nx > 0$.

the $\lim_{n \rightarrow \infty} \sin(n+1)x = 0$.

$$\sin(n+1)x = \sin nx \cos x + \cos nx \sin x$$

Since $\sin x \neq 0, \lim_{n \rightarrow \infty} \sin nx \cos x = \cos x \cdot \lim_{n \rightarrow \infty} \sin nx = 0$.

then we must have $\lim_{n \rightarrow \infty} \cos nx = 0$.

Which contradicts with the fact $\sin^2 x + \cos^2 x = 1$.

Hence, $\lim_{n \rightarrow \infty} \sin nx \neq 0$. i.e. the series is divergent.

$$131. (a). \lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n \leq \lim_{n \rightarrow \infty} (x_n + y_n) \leq \lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n$$

Proof: $\lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n \leq \lim_{n \rightarrow \infty} (x_n + y_n)$

① First we consider the $\lim_{n \rightarrow \infty} x_n = +\infty$, we claim that $\lim_{n \rightarrow \infty} y_n \neq -\infty$, otherwise, the sum of $\lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n$ has no meaning.

It follows that $\lim_{n \rightarrow \infty} y_n = +\infty$ or $(A \in \mathbb{R})$, then $\lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n = +\infty$. Since $\{y_n\}$ not tends to $-\infty$, $\{x_n + y_n\} \rightarrow +\infty$, $\lim_{n \rightarrow \infty} x_n + y_n = +\infty$, the equality holds.

② Then we consider the $\lim_{n \rightarrow \infty} x_n = -\infty$, similarly, we have $\lim_{n \rightarrow \infty} y_n \neq +\infty$. Then the "no less large than mark" holds trivially.

③ If $\lim_{n \rightarrow \infty} y_n = \pm\infty$, similar as ①/② (symmetry).

④ We consider $\lim_{n \rightarrow \infty} x_n = A$, $\lim_{n \rightarrow \infty} y_n = B$ ($A, B \in \mathbb{R}$). $\{x_n\}, \{y_n\}$ is bounded below, hence the $\lim_{n \rightarrow \infty} (x_n + y_n) = C$ or $+\infty$.

If $\lim_{n \rightarrow \infty} (x_n + y_n) = +\infty$, obviously the equation holds.

If $\lim_{n \rightarrow \infty} (x_n + y_n) = C$, we can find a subsequence $\{x_{p_n} + y_{p_n}\}$ of $\{x_n + y_n\}$ s.t. $\lim_{n \rightarrow \infty} (x_{p_n} + y_{p_n}) = C$.

We can claim $\{x_{p_n}\}$ and $\{y_{p_n}\}$ is bounded above.

otherwise $\{x_{p_n} + y_{p_n}\}$ is divergent.

Let $\{x_{p_{n_1}}\}$ a subsequence of $\{x_{p_n}\}$, $\{y_{p_{n_1}}\}$ -

$$\lim_{n \rightarrow \infty} x_n \leq \lim_{n \rightarrow \infty} x_{p_{n_1}} \quad \lim_{n \rightarrow \infty} y_n \leq \lim_{n \rightarrow \infty} y_{p_{n_1}}$$

$$\lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n \leq \lim_{n \rightarrow \infty} (x_{p_{n_1}} + y_{p_{n_1}}) \leq \lim_{n \rightarrow \infty} (x_{p_n} + y_{p_n}) = C = \lim_{n \rightarrow \infty} (x_n + y_n)$$

$$(1) \lim_{n \rightarrow \infty} (x_n + y_n) \leq \lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n$$

$$\lim_{n \rightarrow \infty} y_n = -\lim_{n \rightarrow \infty} (-y_n)$$

$$\text{similar as (1). } \lim_{n \rightarrow \infty} (-y_n) + \lim_{n \rightarrow \infty} (x_n + y_n) \leq \lim_{n \rightarrow \infty} [-y_n + (x_n + y_n)] = \lim_{n \rightarrow \infty} x_n$$

$$\text{i.e. } \lim_{n \rightarrow \infty} (x_n + y_n) \leq \lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n$$

$$\text{eg.: } \{x_n\} = 0, 1, 0, 1, \dots$$

$$\{y_n\} = 2, 0, 2, 0, \dots$$

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} y_n = 0. \quad \lim_{n \rightarrow \infty} (x_n + y_n) = 1, \quad \lim_{n \rightarrow \infty} y_n = 2$$

132. Let $x_n \geq 0$ and $y_n \geq 0$. ($n = 1, 2, \dots$). Verify:

$$(a) \lim_{n \rightarrow \infty} x_n \cdot \lim_{n \rightarrow \infty} y_n \leq \lim_{n \rightarrow \infty} (x_n \cdot y_n) \leq \lim_{n \rightarrow \infty} x_n \cdot \lim_{n \rightarrow \infty} y_n$$

(i) ① First, we consider the situation $\lim_{n \rightarrow \infty} x_n = 0$.

then $\lim_{n \rightarrow \infty} y_n \neq 0$ and $\lim_{n \rightarrow \infty} x_n \cdot \lim_{n \rightarrow \infty} y_n = 0$. @, the inequality holds.

② Then we consider the situation $\lim_{n \rightarrow \infty} x_n = +\infty$.

~~then~~ it follows that $\lim_{n \rightarrow \infty} y_n \neq 0$, $\lim_{n \rightarrow \infty} x_n \neq 0$ and $\lim_{n \rightarrow \infty} y_n = +\infty$.

$\lim_{n \rightarrow \infty} y_n \neq 0$, $\exists \varepsilon_0 > 0$, $\exists N > 0$, $y_n > \varepsilon_0$ for all $n > N$.

$$\lim_{n \rightarrow \infty} (x_n \cdot y_n) = \lim_{n \rightarrow \infty} (x_n \cdot y_n) = +\infty. \quad < \lim$$

③ Finally we consider $\lim_{n \rightarrow \infty} x_n = a$ ($a \in \mathbb{R}_+$). $\lim_{n \rightarrow \infty} y_n = b$ ($b \in \mathbb{R}_+$)

If $\lim_{n \rightarrow \infty} (x_n \cdot y_n) \neq ab$ then the inequality hold

If $\lim_{n \rightarrow \infty} (x_n \cdot y_n) = c \in \mathbb{R}_+$, we can find subsequence $\{x_{p_i} \cdot y_{p_i}\}$.

similar as [131], find $\{x_{p_i}\}$, $\{y_{p_i}\}$

$$\lim_{n \rightarrow \infty} x_n \cdot \lim_{n \rightarrow \infty} y_n \leq \lim_{n \rightarrow \infty} x_{p_i} \cdot \lim_{n \rightarrow \infty} y_{p_i} \leq \lim_{n \rightarrow \infty} (x_{p_i} \cdot y_{p_i}) \leq \lim_{n \rightarrow \infty} (x_n \cdot y_n)$$

⊕ (2) If $\overline{\lim}_{n \rightarrow \infty} y_n = 0$, then $\{y_n\}$ has infinite terms equal to 0.

$$\overline{\lim}_{n \rightarrow \infty} (x_n \cdot y_n) = 0 \quad \text{Else, } \overline{\lim}_{n \rightarrow \infty} y_n = \frac{1}{\overline{\lim}_{n \rightarrow \infty} \frac{1}{y_n}}$$

$$\frac{\overline{\lim}_{n \rightarrow \infty} \frac{1}{y_n} \cdot \overline{\lim}_{n \rightarrow \infty} (x_n \cdot y_n)}{\overline{\lim}_{n \rightarrow \infty} \frac{1}{y_n}} \leq \frac{\overline{\lim}_{n \rightarrow \infty} [\frac{1}{y_n} \cdot (x_n \cdot y_n)]}{\overline{\lim}_{n \rightarrow \infty} \frac{1}{y_n}} = \overline{\lim}_{n \rightarrow \infty} x_n$$

$$(3) \{x_n\} = \frac{1}{2}, 2, \frac{1}{2}, 2, \dots$$

$$\{y_n\} = 3, \frac{1}{2}, 3, \frac{1}{2}, \dots$$

$$\{x_n y_n\} = \frac{3}{2}, 1, \frac{3}{2}, 1, \dots \quad \text{"less than" holds.}$$

133(a) If $\lim_{n \rightarrow \infty} x_n$ exists, then for any sequence y_n ($n=1, 2, \dots$)

$$\neq \overline{\lim}_{n \rightarrow \infty} (x_n + y_n) = \overline{\lim}_{n \rightarrow \infty} x_n + \overline{\lim}_{n \rightarrow \infty} y_n;$$

Proof: If $\overline{\lim}_{n \rightarrow \infty} y_n = \pm \infty$, since $\{x_n\}$ is convergent,

$$\text{then } \overline{\lim}_{n \rightarrow \infty} (y_n + x_n) = \pm \infty.$$

$$\text{If } \overline{\lim}_{n \rightarrow \infty} y_n = b \ (b \in \mathbb{R}),$$

$$\overline{\lim}_{n \rightarrow \infty} (y_n x_n + y_n) \leq \overline{\lim}_{n \rightarrow \infty} x_n + \overline{\lim}_{n \rightarrow \infty} y_n = \overline{\lim}_{n \rightarrow \infty} x_n + \overline{\lim}_{n \rightarrow \infty} y_n \leq \frac{\overline{\lim}_{n \rightarrow \infty} x_n + \overline{\lim}_{n \rightarrow \infty} y_n}{\overline{\lim}_{n \rightarrow \infty} \frac{1}{y_n}}$$

$$\leq \overline{\lim}_{n \rightarrow \infty} (x_n + y_n). \quad \text{Hence, } \overline{\lim}_{n \rightarrow \infty} (x_n + y_n) = \overline{\lim}_{n \rightarrow \infty} x_n + \overline{\lim}_{n \rightarrow \infty} y_n$$

[134] Verify: For sequence $\{x_n\}$, any sequence $\{y_n\}$ has at least one of these ~~equation~~ equality holds

$$(a) \overline{\lim}_{n \rightarrow \infty} (x_n + y_n) = \overline{\lim}_{n \rightarrow \infty} x_n + \overline{\lim}_{n \rightarrow \infty} y_n$$

$$(b) \overline{\lim}_{n \rightarrow \infty} (x_n y_n) = \overline{\lim}_{n \rightarrow \infty} x_n \cdot \overline{\lim}_{n \rightarrow \infty} y_n \quad (x_n \geq 0)$$

then the sequence $\{x_n\}$ is convergent or divergent to $+\infty$

Proof: (1) Assume $\lim_{n \rightarrow \infty} x_n = -\infty$, (b) can't hold.

Now consider (a). If $\overline{\lim}_{n \rightarrow \infty} y_n$, (a) can't hold.

(2) Assume $\{x_n\}$ is divergent and $\lim_{n \rightarrow \infty} x_n$ not exists.

Suppose $\lim_{n \rightarrow \infty} x_n = \alpha$, $\overline{\lim}_{n \rightarrow \infty} x_n = \beta$.

① $\{x_n\}$ doesn't subject to $x_n \geq 0$, so we should only consider (a).

$$\text{Let } y_n = -x_n. \quad \overline{\lim}_{n \rightarrow \infty} (x_n + y_n) = 0.$$

$$\overline{\lim}_{n \rightarrow \infty} x_n + \overline{\lim}_{n \rightarrow \infty} y_n = \overline{\lim}_{n \rightarrow \infty} x_n - \overline{\lim}_{n \rightarrow \infty} x_n = \beta - \alpha. \quad \text{Since } \alpha < \beta, \beta - \alpha \neq 0.$$

② $\{x_n\}$ is nonnegative sequence.

We can find a subsequence $\{x_{p_n}\} \subseteq \{x_n\}$, s.t. $x_{p_n} \rightarrow \alpha \geq 0$.

We let $y_n = \begin{cases} 1, & k = p_n, n = 1, 2, \dots \\ -x_k, & \text{others.} \end{cases}$

$$\overline{\lim}_{n \rightarrow \infty} y_n = 1 \quad \overline{\lim}_{n \rightarrow \infty} (x_n + y_n) = \alpha + 1 = \overline{\lim}_{n \rightarrow \infty} x_n + 1 \quad \overline{\lim}_{n \rightarrow \infty} (x_n y_n) = \alpha.$$

(a) we ~~can~~ have $\alpha + 1 \neq \beta + 1$ (b) we ~~can~~ have $\alpha \neq \beta$.

Hence, these assumptions are false. $\square$.

[135] Verify: If $x_n > 0$ ($n=1, 2, \dots$) and $\overline{\lim}_{n \rightarrow \infty} x_n \cdot \overline{\lim}_{n \rightarrow \infty} \frac{1}{x_n} = 1$

then $\{x_n\}$ is convergent.

Proof: ① If $\lim_{n \rightarrow \infty} x_n = \pm \infty$,

then $\overline{\lim}_{n \rightarrow \infty} x_n = \pm \infty$, the equality can't hold.

② If $\lim_{n \rightarrow \infty} x_n$ has no meaning.

We ~~not~~ assume $\overline{\lim}_{n \rightarrow \infty} \frac{1}{x_n} = \alpha$ ($\alpha \in \mathbb{R}$). otherwise $\overline{\lim}_{n \rightarrow \infty} \frac{1}{x_n} = \pm \infty$,

the equality cannot hold. since $x_n > 0$, we have,

$$\overline{\lim}_{n \rightarrow \infty} \frac{1}{x_n} = \frac{1}{\lim_{n \rightarrow \infty} x_n} \Rightarrow \overline{\lim}_{n \rightarrow \infty} \frac{1}{x_n} \cdot \lim_{n \rightarrow \infty} x_n = 1.$$

Hence, $\overline{\lim}_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} x_n \neq \pm \infty$, i.e. $\{x_n\}$ is convergent.

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Verify: $\lim_{x \rightarrow x_0} f(x)$ exists. $\Leftrightarrow (\lim_{x \rightarrow x_0^+} f(x)$ and $\lim_{x \rightarrow x_0^-} f(x))$ exists. $\wedge \lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0^-} f(x)$

proof: " $\Rightarrow$ " Suppose $\lim_{x \rightarrow x_0} f(x) = A$.

$\forall \varepsilon > 0, \exists \delta > 0, 0 < |x - x_0| < \delta$, we have $|f(x) - A| < \varepsilon$.

since $x_0 - \delta < x < x_0 + \delta$,

~~we can~~ the $x_0 < x < x_0 + \delta$, we have $|f(x) - A| < \varepsilon$.

i.e. $\lim_{x \rightarrow x_0^+} f(x) = A$. similarly, $x_0 - \delta < x < x_0$, $\lim_{x \rightarrow x_0^-} f(x) = A$.

" $\Leftarrow$ " $\forall \varepsilon > 0, \exists \delta_1 > 0, x_0 - \delta_1 < x < x_0, |f(x) - A| < \varepsilon$.

$\forall \varepsilon > 0, \exists \delta_2 > 0, x_0 < x < x_0 + \delta_2, |f(x) - A| < \varepsilon$.

(We suppose $\lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0^-} f(x) = A$)

Let $\delta = \max \{ \delta_1, \delta_2 \}$ $x_0 - \delta < x < x_0 + \delta$

i.e. $0 < |x - x_0| < \delta$, $|f(x) - A| < \varepsilon$.

i.e. $\lim_{x \rightarrow x_0} f(x) = A$.

[138] Verify: If the sequence $\{x_n\}$ converges, then the sequence

of arithmetic mean of $\{x_n\}$ converges and $\lim_{n \rightarrow \infty} \frac{x_1 + x_2 + \dots + x_n}{n} = \lim_{n \rightarrow \infty} x_n$.

The converse sentence doesn't hold, give example.

Proof: Suppose that $\lim_{n \rightarrow \infty} \frac{x_1 + x_2 + \dots + x_n}{n} = \lim_{n \rightarrow \infty} x_n = a$.

1° if $a = 0$. $\forall \varepsilon > 0, \exists N \in \mathbb{N}, |x_n| < \frac{\varepsilon}{2}$ for any $n > N$.

We can also find $N_2 \in \mathbb{N}, \left| \frac{x_1 + \dots + x_{N_2}}{N_2} \right| < \frac{\varepsilon}{2}$ for any $N_2 > N$.

Let $N = \max \{ N_1, N_2 \}$, for $n > N$, we have

$$\left| \frac{x_1 + \dots + x_n}{n} \right| \leq \left| \frac{x_1 + \dots + x_{N_2}}{n} \right| + \left| \frac{x_{N_2+1} + \dots + x_n}{n} \right| \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

$$\text{i.e. } \lim_{n \rightarrow \infty} \frac{x_1 + x_2 + \dots + x_n}{n} = 0$$

$\circ$ if $a \neq 0$, similarly, $\forall \varepsilon > 0$,

We can find $N_1 \in \mathbb{N}$, s.t. $|x_n - a| < \frac{\varepsilon}{2}$ for any $n \in \mathbb{N}$,

$N_2 \in \mathbb{N}$, s.t. $\left| \frac{(x_1 - a) + (x_2 - a) + \dots + (x_N - a)}{n} \right| < \frac{\varepsilon}{2}$ for any $n > N_2$.

$N = \max\{N_1, N_2\}$ for any $n > N$.

$$\left| \frac{x_1 + x_2 + \dots + x_n}{n} - a \right| = \left| \frac{(x_1 - a) + \dots + (x_n - a)}{n} \right| < \varepsilon.$$

$$\lim_{n \rightarrow \infty} x_n = a.$$

e.g. $x_n = (-1)^n$. $\lim_{n \rightarrow \infty} \frac{x_1 + \dots + x_n}{n} = 0$. $\{x_n\}$ is divergent

[143] (a), $y_{n+1} > y_n$ (b) $\lim_{n \rightarrow \infty} y_n = +\infty$; (c) $\lim_{n \rightarrow \infty} \frac{x_{n+1} - x_n}{y_{n+1} - y_n}$ exists.

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{x_n}{y_n} = \lim_{n \rightarrow \infty} \frac{x_{n+1} - x_n}{y_{n+1} - y_n}$$

$$\text{Let } \underline{x}_1 = x_1, \underline{x}_2 = x_2 - x_1, \underline{x}_3 = x_3 - x_2, \dots, \underline{x}_n = x_n - x_{n-1}$$

$$y_1 = y_1, y_2 = y_2 - y_1, y_3 = y_3 - y_2, \dots, y_n = y_n - y_{n-1}$$

$$\text{then } x_{n+1} = \underline{x}_{n+1} + \underline{x}_n + \dots + \underline{x}_2 + \underline{x}_1, y_{n+1} = y_{n+1} + y_n + \dots + y_2 + y_1$$

$$\text{Now we should prove } \lim_{n \rightarrow \infty} \frac{\underline{x}_1 + \dots + \underline{x}_n}{y_1 + \dots + y_n} = \lim_{n \rightarrow \infty} \frac{\underline{x}_n}{y_n}$$

$$\text{Let } \lim_{n \rightarrow \infty} \frac{\underline{x}_n}{y_n} = a. \text{ i.e. } \frac{\underline{x}_n}{y_n} = \alpha + a, \alpha \rightarrow 0.$$

$$\underline{x}_n = \alpha y_n + a y_n, \forall \varepsilon > 0, \exists N \in \mathbb{N}, \text{ s.t. } |\alpha| < \frac{\varepsilon}{2}, \text{ for any } n > N$$

$$\left| \frac{\underline{x}_1 + \dots + \underline{x}_n}{y_1 + \dots + y_n} - a \right| = \left| \frac{(\underline{x}_1 - a y_1) + (\underline{x}_2 - a y_2) + \dots + (\underline{x}_n - a y_n)}{y_1 + \dots + y_n} \right|$$

$$\leq \left| \frac{(\underline{x}_1 - a y_1) + (\underline{x}_2 - a y_2) + \dots + (\underline{x}_n - a y_n)}{y_1 + \dots + y_n} \right| + \frac{\varepsilon}{2}$$

$$\text{Since } \lim_{n \rightarrow \infty} (y_1 + y_2 + \dots + y_n) = +\infty, \text{ then } \exists N_2 \in \mathbb{N}, \text{ s.t. } \left| \frac{\underline{x}_1 - a y_1 + \dots + \underline{x}_n - a y_n}{y_1 + \dots + y_n} \right| < \frac{\varepsilon}{2}.$$

$$\left| \frac{\underline{x}_1 + \dots + \underline{x}_n}{y_1 + \dots + y_n} - a \right| < \varepsilon: \text{ thus, } \lim_{n \rightarrow \infty} \frac{\underline{x}_1 + \dots + \underline{x}_n}{y_1 + \dots + y_n} = a.$$

$$\text{Let } N_2 = \max\{N, N_1\}, \text{ for any } n > N_2, \text{ i.e. } \lim_{n \rightarrow \infty} \frac{x_n}{y_n} = \lim_{n \rightarrow \infty} \frac{x_{n+1} - x_n}{y_{n+1} - y_n} = a.$$

[144] (b) $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = \lim_{n \rightarrow \infty} \ln \cdot e^{\frac{\ln n}{n}} = \ln \left(\lim_{n \rightarrow \infty} \sqrt[n]{n} \right) = \ln 1 = 0$.
 (Proof $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$. [65])

[149] $x_0 > 0$, $x_{n+1} = \frac{1}{2} \left(x_n + \frac{1}{x_n} \right)$. Verify: $\lim_{n \rightarrow \infty} x_n = 1$.

$$x_{n+1} \geq \frac{1}{2} \cdot 2 \sqrt{x_n \cdot \frac{1}{x_n}} = 1$$

$$|x_{n+1} - 1| = \frac{1}{2} \left| \frac{x_n^2 - 2x_n + 1}{x_n} \right| = \frac{1}{2} \cdot \frac{|x_n - 1|^2}{|x_n|} < \frac{x_n - 1}{2x_n} \cdot |x_n - 1| = \frac{1}{2} |x_n - 1|$$

$$|x_n - 1| \leq \frac{1}{2} |x_{n-1} - 1| < \frac{1}{2^2} |x_{n-2} - 1| < \dots < \frac{1}{2^n} |x_0 - 1|.$$

Since $\lim_{n \rightarrow \infty} \frac{1}{2^n} = 0$, $0 < |x_n - 1| < \frac{1}{2^n} |x_0 - 1|$

By squeeze theorem, $\lim_{n \rightarrow \infty} |x_n - 1| = 0$, i.e. $\lim_{n \rightarrow \infty} x_n = 1$.

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$$[15b]. x_1 = a, y_1 = b. x_{n+1} = \sqrt{x_n y_n} \quad y_{n+1} = \frac{x_n + y_n}{2}$$

$$\mu(a, b) = \lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} y_n$$

$$1^\circ a \cdot b = 0. \text{ then } x_{n+1} = 0. \quad y_{n+1} = \frac{y_n}{2} \quad (n \geq 1)$$

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} y_n = 0.$$

$$2^\circ a, b \neq 0, \text{ then } a > 0, b > 0.$$

$$\text{we can find } a = x_1 \leq x_2 \leq x_3 \leq \dots \leq x_n \leq y_n \leq \dots \leq y_3 \leq y_2 \leq y_1 = b.$$

$$\{x_n\} \text{ is nondecreasing and bounded above. let } \lim_{n \rightarrow \infty} x_n = A.$$

$$\text{Similarly, let } \lim_{n \rightarrow \infty} y_n = B.$$

$$\text{When } n \rightarrow \infty, A = B = \mu(a, b)$$

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[b] [b]

$$(a) \ o(o(f(x))) = o(f(x)) \cdot \alpha_1(x) \quad (\lim \alpha_1(x) = 0)$$

$$= f(x) \cdot \alpha_2(x) \cdot \alpha_1(x) \quad (\text{For } \lim \alpha_2(x) = 0)$$

Since ~~that~~ $\lim \alpha_1(x) \cdot \alpha_2(x) = \lim \alpha_1(x) \cdot \lim \alpha_2(x) = 0$.

We have $o(o(f(x))) = f(x) \cdot (\alpha_2(x) \cdot \alpha_1(x)) = o(f(x))$

(b) $\lim \frac{o(o(f(x)))}{o(f(x))} = c$.

i.e. $\frac{o(o(f(x)))}{o(f(x))} = c + \alpha_1(x) \quad (\alpha_1(x) \rightarrow 0)$

We suppose $o(f(x)) = \alpha_2(x) \cdot f(x) \quad (\alpha_2(x) \rightarrow 0)$

$$\therefore o(o(f(x))) = [\alpha_2(x) f(x)] (c + \alpha_1(x))$$

$$= f(x) \cdot [\alpha_2(x) \cdot c + \alpha_1(x) \alpha_2(x)]$$

Since ~~that~~ $\alpha_2(x) \cdot c + \alpha_1(x) \alpha_2(x) \rightarrow 0$.

then $o(o(f(x))) = o(f(x))$

(c) Similar as (b), we have

$$o(o(f(x))) = o(f(x)) \cdot \alpha_1(x) = (c + \alpha_2(x)) \cdot f(x) \cdot \alpha_1(x)$$

$$= f(x) [\alpha_1(x) c + \alpha_2(x) \alpha_1(x)] \quad (\text{For } \alpha_1(x) \rightarrow 0, \alpha_2(x) \rightarrow 0, c \in \mathbb{R} \setminus \{0\})$$

Similarly, $o(o(f(x))) = o(f(x))$

(d) $\lim \frac{o(o(f(x)))}{o(f(x))} = c_1 (c_1 \neq 0, c_1 \in \mathbb{R})$, $\lim \frac{o(f(x))}{f(x)} = c_2 (c_2 \in \mathbb{R} \setminus \{0\})$

$$o(o(f(x))) = [c_1 + \alpha_1(x)] \cdot o(f(x))$$

$$\therefore \lim \frac{o(o(f(x)))}{f(x)} = \frac{[c_1 + \alpha_1(x)] \cdot o(f(x))}{f(x)} = \lim \frac{o(f(x)) \cdot c_1}{f(x)} + 0$$

$$\therefore o(o(f(x))) = o(f(x)) \quad = c_1 \cdot c_2 \cdot f(x) \in (\mathbb{R} \setminus \{0\})$$

$$(e) O(f(x)) + o(f(x)) = O(f(x))$$

$$\lim_{x \rightarrow 0} \frac{O(f(x)) + o(f(x))}{f(x)} = \lim_{x \rightarrow 0} \frac{O(f(x))}{f(x)} + \lim_{x \rightarrow 0} \frac{o(f(x))}{f(x)} = c_1 + 0 = c_1 \quad (c_1 \in \mathbb{R})$$

hence, the equality holds.

Ex 3) (a) principal part: $2x$

order: 1

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x} = \lim_{x \rightarrow 0} \frac{(1+x) - (1-x)}{(\sqrt{1+x} + \sqrt{1-x})x} = \lim_{x \rightarrow 0} \frac{2x}{(\sqrt{1+x} + \sqrt{1-x})x} = 1$$

principal part: x

order: 1

$$(c) (1-2x)^{\frac{1}{2}} = 1 - x - \frac{1}{8}(2x)^2 + o(x^2)$$

order 2

$$(1-3x)^{\frac{1}{3}} = 1 - x - \frac{1}{9}(\frac{1}{3}x)^2 + o(x^2)$$

principal part: $\frac{1}{2}x^2$

$$\lim_{x \rightarrow 0} \frac{(1-2x)^{\frac{1}{2}} - (1-3x)^{\frac{1}{3}}}{x^2} = \frac{\frac{1}{2}x^2}{x^2} = \frac{1}{2}$$

$$(d) \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x} = \lim_{x \rightarrow 0} \frac{\sin x - \sin x \cos x}{\cos x \cdot x} = \lim_{x \rightarrow 0} \frac{\sin x (1 - \cos x)}{\cos x \cdot x}$$

$$= \lim_{x \rightarrow 0} \frac{x \cdot \frac{1}{2}x^2}{x} = \lim_{x \rightarrow 0} \frac{\frac{1}{2}x^3}{x}$$

principal part: $\frac{1}{2}x^3$

order: 3

$$[65] \lim_{x \rightarrow 1} \frac{(x-1)^2(x+1)}{(x-1)^2} = 3.$$

principal part : $3(x-1)^2$

order : 2.

$$(b) \sqrt[3]{1-\sqrt{x}} = \sqrt[3]{\frac{(1-\sqrt{x})(1+\sqrt{x})}{1+\sqrt{x}}} = \sqrt[3]{\frac{(1-x)}{1+\sqrt{x}}} = \frac{(1-x)^{\frac{1}{3}}}{\sqrt[3]{1+\sqrt{x}}}$$

$$\lim_{x \rightarrow 1} \frac{\frac{(1-x)^{\frac{1}{3}}}{\sqrt[3]{1+\sqrt{x}}}}{(x-1)^{\frac{1}{3}}} = \frac{1}{\sqrt[3]{2}}$$

principal part : $\frac{1}{\sqrt[3]{2}} (1-x)^{\frac{1}{3}}$ order : $\frac{1}{3}$.

$$(c) \lim_{x \rightarrow 1} \frac{\ln x}{x-1} \quad \frac{\ln x = t}{t \rightarrow 0} \quad \lim_{t \rightarrow 0} \frac{t}{e^t - 1} = 1.$$

principal part : $(x-1)$

order : 1

$$(d) \lim_{x \rightarrow 1} \frac{e^x - e}{x-1} = \lim_{x \rightarrow 1} \frac{e(e^{x-1} - 1)}{x-1} = e.$$

principal part : $e(x-1)$ order : e.

$$[65] \frac{x+1}{x^4+1} = \frac{(\frac{1}{x^3}) + (\frac{1}{x^4})}{1 + (\frac{1}{x^4})} = 1 + \frac{\frac{1}{x^3} - 1}{1 + \frac{1}{x^4}}$$

$$\lim_{x \rightarrow \infty} \frac{\frac{x+1}{x^4+1}}{(\frac{1}{x})^3} = \lim_{x \rightarrow \infty} \frac{x^4+x^3}{x^4+1} = 1$$

principal part : $(\frac{1}{x})^3$ order : 3.

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$$(b) \lim_{x \rightarrow \infty} \frac{\sqrt{x+1} - \sqrt{x}}{(\frac{1}{x})^{\frac{1}{2}}} = \lim_{x \rightarrow \infty} \frac{1}{(\sqrt{x} + \sqrt{x+1})(\frac{1}{x})^{\frac{1}{2}}} = \lim_{x \rightarrow \infty} \frac{1}{1 + \sqrt{1 + \frac{1}{x}}} = \frac{1}{2}$$

principal part: $\frac{1}{2}(\frac{1}{x})^{\frac{1}{2}}$ order: $\frac{1}{2}$.

$$(c) \lim_{x \rightarrow \infty} \frac{\sqrt{x+2} - \sqrt{x+1} - (\sqrt{x+1} - \sqrt{x})}{(\frac{1}{x})^{\frac{3}{2}}}$$

$$\begin{aligned} f(x) = \sqrt{x+2} + \sqrt{x} - 2\sqrt{x+1} &= \frac{(\sqrt{x+2} + \sqrt{x} - 2\sqrt{x+1})(\sqrt{x+2} + \sqrt{x} + 2\sqrt{x+1})}{\sqrt{x+2} + \sqrt{x} + 2\sqrt{x+1}} \\ &= \frac{(\sqrt{x+2} + \sqrt{x})^2 - 4(x+1)}{\sqrt{x+2} + \sqrt{x} + 2\sqrt{x+1}} \\ &= \frac{2\sqrt{x(x+2)} - 2(x+1)}{\sqrt{x+2} + \sqrt{x} + 2\sqrt{x+1}} = \frac{-2[\sqrt{x(x+2)} + (x+1)]}{(\sqrt{x+2} + \sqrt{x} + 2\sqrt{x+1})(\sqrt{x+2} + \sqrt{x} + 2\sqrt{x+1})} \\ &= \frac{-2}{(x+1 - \sqrt{x(x+2)})(\sqrt{x+2} + \sqrt{x} + 2\sqrt{x+1})} \end{aligned}$$

$$\lim_{x \rightarrow \infty} \frac{f(x)}{(\frac{1}{x})^{\frac{3}{2}}} = \lim_{x \rightarrow \infty} \frac{-2}{(\frac{x+1 - \sqrt{x(x+2)}}{x})(\frac{\sqrt{x+2} + \sqrt{x} + 2\sqrt{x+1}}{\sqrt{x}})}$$

$$= \frac{-2}{(1+1)(1+1+2)} = -\frac{1}{4}$$

per-principle part: $-\frac{1}{4}(\frac{1}{x})^{\frac{3}{2}}$ order: $\frac{3}{2}$

$$(d) \lim_{x \rightarrow \infty} \frac{\frac{1}{x} \sin x}{(\frac{1}{x})^2} \stackrel{\frac{1}{x} = t}{=} \lim_{t \rightarrow 0} \frac{\sin t}{t} = 1$$

principle. part $(\frac{1}{x})^2$ order: 2

[658]

$$(a) \lim_{x \rightarrow 1} \frac{x^2}{x^2 - 1} = \lim_{x \rightarrow 1} \frac{x^2}{(x-1)(x+1)} = \lim_{x \rightarrow 1} \frac{x}{x+1} = \frac{1}{2} \quad \frac{1}{2} \left(\frac{1}{x+1} \right), 1.$$

$$(b) \lim_{x \rightarrow 1} \frac{\sqrt{1+x}}{\sqrt{1-x} \cdot \left(\frac{1}{1-x} \right)^{\frac{1}{2}}} = \lim_{x \rightarrow 1} \sqrt{1+x} = \sqrt{2} \quad \sqrt{2} \cdot \left(\frac{1}{x+1} \right)^{\frac{1}{2}}, \frac{1}{2}.$$

$$(c) \lim_{x \rightarrow 1} \frac{x}{\sqrt[3]{(1-x)(x^2+x+1)} \cdot \left(\frac{1}{1-x} \right)^{\frac{1}{3}}} = \lim_{x \rightarrow 1} \frac{x}{\sqrt[3]{x^2+x+1}} = \frac{1}{\sqrt[3]{3}} \quad \frac{1}{\sqrt[3]{3}} \left(\frac{1}{x+1} \right)^{\frac{1}{3}}, \frac{1}{3}.$$

$$(d) \lim_{x \rightarrow 1} \frac{x-1}{\sin \pi x} = \lim_{x \rightarrow 1} \frac{x-1}{\sin [\pi(x-1) + \pi]} = \lim_{x \rightarrow 1} \frac{x-1}{-\sin \pi(x-1)} = -\frac{1}{\pi} \\ -\frac{1}{\pi} \cdot \frac{1}{x-1}, \text{ order 1.}$$

$$(e) \lim_{x \rightarrow 1} \frac{O(\ln x)}{(1-x)^2 \cdot \frac{1}{x-1}} = \lim_{x \rightarrow 1} \frac{\ln x}{x-1} = 1.$$

principle part $x-1$. order 1.

[674] (a). $f(x) = ax+b$.

$\forall \varepsilon > 0$. Let $\delta = \frac{\varepsilon}{a}$, For any $0 < |x - x_0| < \delta$.

$$\text{sit. } |f(x) - f(x_0)| = |ax - ax_0| < a \cdot \frac{\varepsilon}{a} = \varepsilon$$

Since $x_0 \in \mathbb{R}$ and x_0 is arbitrary.

$f(x) = ax+b$ is continuous over $\mathbb{R}$.